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INFORMATION RETRIEVAL

INTERNAL ASSESSMENT -I

Unit I- Introduction to Information Retrieval

Ques 1. Define Information Retrieval and explain its goals and history.

Information Retrieval is the process of searching for and obtaining relevant information from a large collection of data based on a user's query. It mainly deals with retrieving information from unstructured or semi-structured data sources such as documents, web pages, emails, and multimedia content.

In simple terms, Information Retrieval helps users find useful and relevant information from massive datasets quickly and efficiently. Unlike traditional databases that return exact matches, IR systems focus on providing the most relevant results, even if they are not an exact match to the query.

Goals of Information Retrieval:

The main objective of Information Retrieval is to satisfy the user's information need in the best possible way. Some key goals are:

1. **Relevance:**
   The system should return results that are closely related to the user's query. Irrelevant information should be minimized.
2. **Efficiency:**
   Retrieval should be fast, even when dealing with very large datasets like the internet.
3. **Accuracy:**

The system should correctly interpret the user's query and provide precise results.

4. **Scalability:**
   IR systems should be able to handle increasing amounts of data without performance issues.
5. **User Satisfaction:**
   The final aim is to ensure that the user finds the required information easily and effectively.
6. **Ranking of Results:**
   Results should be displayed in an order where the most relevant information appears first.

History of Information Retrieval:

The development of Information Retrieval has evolved over time along with advancements in technology:

- **1940s–1950s (Early Concepts):**
  The idea of retrieving information began with manual indexing and classification systems used in libraries. During this time, the focus was on organizing documents for easy access.
- **1960s (Birth of IR Systems):**
  The first computerized IR systems were developed. Researchers started using automatic indexing and keyword-based search techniques.
- **1970s–1980s (Model Development):**
  Important models like the Boolean model, vector space model, and probabilistic model were introduced. These models improved how documents were matched with queries.
- **1990s (Web Era Begins):**
  With the rise of the internet, Information Retrieval became more important. Search engines like Yahoo and early versions of Google started using advanced ranking algorithms to retrieve web pages.
- **2000s–Present (Modern IR):**

Modern IR systems use machine learning, natural language processing, and artificial intelligence to improve search results. Today's systems understand user intent better and provide more personalized and accurate results.

Information Retrieval plays a crucial role in today's digital world, where vast amounts of data are generated every second. From search engines to digital libraries, IR systems help users access the right information quickly, making them an essential part of modern computing.

Ques 2. Explain the impact of the World Wide Web on Information Retrieval systems.

The introduction of the World Wide Web has had a significant impact on the development and functioning of Information Retrieval (IR) systems. Before the Web, IR systems mainly dealt with small, structured, and controlled datasets such as library records. However, the Web completely transformed the scale, complexity, and approach of information retrieval

1. <u>Massive Growth of Information:</u>
The Web led to an enormous increase in the amount of available data. Millions of websites, documents, images, and videos are continuously being added. This forced IR systems to evolve in order to handle large-scale data efficiently.

2. <u>Shift from Structured to Unstructured Data:</u>
Earlier systems mostly worked with structured databases. With the Web, most data became unstructured (like web pages and blogs). IR systems had to develop new techniques to process and retrieve meaningful information from such data.

3. <u>Development of Search Engines:</u>
The rise of the Web led to the creation of powerful search engines like Google and Yahoo. These systems use advanced algorithms to index and rank billions of web pages, making information easily accessible to users.

4. <u>Introduction of Ranking Algorithms:</u>
Due to the huge number of results, simply retrieving documents was not enough. IR systems began using ranking techniques to display the most relevant results first. For example, Google introduced the PageRank algorithm, which ranks pages based on their importance and links.

5. <u>Increased Focus on Relevance and Personalization:</u>
With diverse users and queries, IR systems started focusing more on understanding user intent. Modern systems provide personalized results based on search history, location, and preferences.

6. <u>Real-Time Information Retrieval:</u>
The Web enabled access to real-time and continuously updated information such as news, weather, and social media updates. IR systems now need to index and retrieve fresh content quickly.

7. <u>Multimedia Retrieval:</u>
The Web includes not only text but also images, audio, and videos. This led to the development of multimedia information retrieval systems that can search across different types of content.

8. <u>Challenges Introduced by the Web:</u>
While the Web improved access to information, it also introduced challenges such as:

- Information overload
- Presence of irrelevant or low-quality data
- Duplicate content
- Spam and misleading information

IR systems had to develop filtering and ranking methods to deal with these issues.

The World Wide Web has revolutionized Information Retrieval by increasing the scale, diversity, and complexity of data. It has led to the development of advanced search engines, improved ranking techniques, and more user-focused retrieval systems. Today, IR systems are an essential part of navigating the vast information available on the Web.

Ques 3. Describe the Information Retrieval process and its main components.

The Information Retrieval process refers to the series of steps through which a system takes a user's query, searches a large collection of data, and returns the most relevant results. The main aim of this process is to match the user's information need with the most appropriate documents available in the system.

Steps in the Information Retrieval Process:

1. **Document Collection:**
   First, a large number of documents (such as web pages, articles, emails, etc.) are collected and stored in the system. This collection is also called a corpus.
2. **Document Processing (Indexing):**
   The collected documents are processed to make searching easier. Important words (keywords) are extracted, and an index is created. This step helps in quickly locating documents during a search.
3. **User Query Input:**
   The user enters a query in the form of keywords or a question. The system interprets this query to understand what the user is looking for.
4. **Query Processing:**
   The query is processed in a similar way to documents. It may involve removing unnecessary words, correcting spelling, or expanding the query with related terms.
5. **Matching and Retrieval:**
   The system compares the processed query with the indexed documents to find matches. It identifies documents that are relevant to the query.
6. **Ranking of Results:**
   The retrieved documents are ranked based on their relevance. The most useful and important results are placed at the top.
7. **Display of Results:**
   Finally, the system presents the results to the user in an organized format, usually as a list.

Main Components of an Information Retrieval System:

1. **Document Collection (Corpus):**
   This is the database of all documents from which information is retrieved.
2. **Indexing Module:**
   It processes documents and creates an index for fast searching.
3. **Query Processor:**
   It analyzes and processes the user's query to improve accuracy.

4. **<u>Matching Mechanism:</u>**
   This component compares the query with stored documents to find relevant matches.
5. **<u>Ranking Algorithm:</u>**
   It orders the retrieved documents based on relevance, ensuring better results appear first.
6. **<u>User Interface:</u>**
   It allows users to enter queries and view results easily.

The Information Retrieval process is a systematic method that transforms a user's query into meaningful results. Its components work together to ensure that users can quickly and efficiently find the information they need from large datasets.

Ques 4. Compare structured, semi-structured, and unstructured text in Information Retrieval.

In Information Retrieval (IR), data can exist in different formats depending on how it is organized. The three main types are structured, semi-structured, and unstructured text. Understanding these types is important because IR systems use different techniques to process and retrieve information from each.

1. <u>Structured Text</u>:
Structured text refers to data that is highly organized and follows a fixed format. It is usually stored in tables with clearly defined fields such as rows and columns. • **Examples:** Databases, spreadsheets, student records

- **Characteristics:**
  - Well-defined schema (fixed structure) ○ Easy to search and retrieve ○ Data is stored in an organized manner
- **In IR:**
  Retrieval is fast and accurate because the system knows exactly where to look for specific information.

2. <u>Semi-Structured Text:</u>

Semi-structured text does not follow a strict table format but still contains some organizational elements such as tags or markers.

- **Examples:** XML files, HTML web pages, JSON data
- **Characteristics:**
  - No rigid schema, but some structure exists ○ Uses tags or labels to organize data ○ Flexible and easier to modify than structured data
- **In IR:**
  Retrieval is moderately complex. Systems use tags and metadata to locate relevant information.

3. <u>Unstructured Text:</u>

Unstructured text is the most common form of data and does not have any predefined format or organization.

- **Examples:** Text documents, emails, social media posts, articles
- **Characteristics:**
  - No fixed structure
  - Difficult to organize and analyze
  - Requires advanced techniques like keyword extraction and natural language processing
- **In IR:**
  Retrieval is more challenging. Systems rely on algorithms to understand meaning and context.

Comparison Table:

Feature	**Structured Text**	**Semi-Structured Text**	**Unstructured Text**
Format	Fixed and organized	Partially organized	No fixed format
Schema	Strict schema	Flexible schema	No schema

Examples	Databases, tables	XML, HTML	Emails, documents
Ease of Retrieval	Easy and fast	Moderate	Difficult
Processing Complexity	Low	Medium	High

Structured, semi-structured, and unstructured texts differ mainly in how data is organized. While structured data is easy to retrieve, unstructured data requires more advanced IR techniques. Modern Information Retrieval systems are designed to handle all three types efficiently, especially the vast amount of unstructured data available today.

Unit II – Basic Retrieval Models

Ques 1. Explain ranked retrieval and discuss different text similarity metrics.

Ranked retrieval is a technique in Information Retrieval (IR) where documents are retrieved and **ordered based on their relevance** to a user's query. Unlike Boolean retrieval (which gives only exact matches), ranked retrieval assigns a **score** to each document and displays results in decreasing order of relevance.

This approach is widely used in modern search systems like Google, where the most relevant pages appear at the top.

Key Idea:
Each document $d$is assigned a score with respect to query q:

$$Score(q, d)$$

Higher score ⇒ More relevant document

Text Similarity Metrics:

Text similarity metrics are used to compute how similar a document is to a query. These are essential for ranking documents.

1. <u>Cosine Similarity:</u>

Cosine similarity measures the cosine of the angle between two vectors (query and document).

$$\cos(\theta) = \frac{\vec{q} \cdot \vec{d}}{\| \vec{q} \| \ \| \vec{d} \|}$$

- Value ranges from **0 to 1**
- 1 means identical, 0 means no similarity
- Works well for text data because it ignores document length

2. <u>Jaccard Similarity:</u>

It measures similarity between two sets of terms.

$$J(A, B) = \frac{| A \cap B |}{| A \cup B |}$$

- A= set of query terms
- B= set of document terms
- Value ranges from **0 to 1**

3. <u>Euclidean Distance:</u>

It calculates the distance between two vectors.

$$d(q, d) = \sqrt{\sum_{i=1}^{n} (q_i - d_i)^2}$$

- Smaller distance ⇒ Higher similarity

- Larger distance ⇒ Lower similarity

4. <u>TF-IDF (Term Frequency–Inverse Document Frequency):</u>

TF-IDF gives importance to words in a document.

- **Term Frequency (TF):**

$$TF(t, d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total terms in document}}$$

- **Inverse Document Frequency (IDF):**

$$IDF(t) = \log\left(\frac{N}{df(t)}\right)$$

- N= total number of documents
- $df(t)$= number of documents containing term t
- **TF-IDF:**

$$TF\text{-}IDF = TF \times IDF$$

Higher value ⇒ more important word

5. <u>Dice Coefficient:</u>

It is another similarity measure based on common terms.

$$Dice(A, B) = \frac{2 \mid A \cap B \mid}{\mid A \mid + \mid B \mid}$$

- Gives more weight to overlapping terms

Ranked retrieval improves search quality by ordering documents according to relevance. Text similarity metrics like cosine similarity, Jaccard similarity, Euclidean

distance, and TF-IDF help in calculating this relevance. These techniques form the foundation of modern Information Retrieval systems.

Ques 2. Describe TF-IDF weighting and explain the importance of term frequency and inverse document frequency.

<u>TF-IDF (Term Frequency–Inverse Document Frequency):</u>

TF-IDF is a numerical weighting technique used in Information Retrieval (IR) to evaluate how important a word (term) is in a document relative to a collection of documents (corpus).

It helps in identifying **important words** in a document while reducing the impact of commonly used words like “the”, “is”, “and”, etc.

<u>Components of TF-IDF</u>

TF-IDF consists of two main parts:

1. <u>Term Frequency (TF):</u>

Term Frequency measures how often a term appears in a document.

$$TF(t, d) = \frac{f(t, d)}{\sum_{t'} f(t', d)}$$

Where:

- $f(t, d)$= frequency of term $t$in document d
- Denominator = total number of terms in document **<u>Importance of TF:</u>**
- Higher TF means the term is more important in that document
- Helps identify the main topic of the document
- However, it alone is not enough because common words may also have high frequency

2. <u>Inverse Document Frequency (IDF):</u>

IDF measures how important a term is across the entire corpus.

$$IDF(t) = \log\left(\frac{N}{df(t)}\right)$$

Where:

- N= total number of documents
- $df(t)$= number of documents containing term t **<u>Importance of IDF:</u>**
- Rare terms get **higher IDF values**
- Common terms get **lower IDF values**
- Helps reduce the importance of frequently used words

<u>TF-IDF Formula</u>

The combined TF-IDF weight is calculated as:

$$TF\text{-}IDF(t, d) = TF(t, d) \times IDF(t)$$

<u>Significance of TF-IDF in Information Retrieval</u>

- Helps in **ranking documents** based on relevance
- Highlights **important and unique terms**
- Reduces the impact of **common or stop words**
- Widely used in search engines and text mining systems • Improves accuracy of retrieval results

Example : If a word appears many times in one document (high TF) but appears in very few documents overall (high IDF), then it is considered very important for that document.

TF-IDF is a powerful weighting technique that combines term frequency and inverse document frequency to determine the importance of words. It plays a

crucial role in improving the performance of Information Retrieval systems by ensuring that relevant documents are ranked higher.

Ques 3. Explain the vector space model and how cosine similarity is used to measure document similarity.

The Vector Space Model is a widely used model in Information Retrieval (IR) where documents and queries are represented as **vectors in a multi-dimensional space**. Each dimension corresponds to a unique term (word) in the document collection.

In this model, a document is represented as:

$$d = (w_1, w_2, w_3, \dots, w_n)$$

where $w_i$represents the weight (usually TF-IDF) of the i^{th}term.

Similarly, a query is also represented as a vector:

$$q = (w_1, w_2, w_3, \dots, w_n)$$

How Vector Space Model Works:

1. Documents and queries are converted into vectors
2. Each term is assigned a weight (commonly using TF-IDF)
3. The similarity between the query and each document is calculated 4.

   Documents are ranked based on similarity scores

<u>Cosine Similarity:</u>

Cosine similarity is the most commonly used method to measure similarity in the Vector Space Model. It calculates the **cosine of the angle** between the document vector and the query vector.

$$\cos(\theta) = \frac{\sum_{i=1}^{n} q_i d_i}{\sqrt{\sum_{i=1}^{n} q_i^2} \sqrt{\sum_{i=1}^{n} d_{i2}}}$$

Interpretation of Cosine Similarity:

- Value ranges from **0 to 1**
- **1** → Document and query are identical (maximum similarity)
- **0** → No similarity
- Smaller angle between vectors ⇒ Higher similarity

Why Cosine Similarity is Used:

- It is independent of document length
- Works well with high-dimensional text data
- Efficient and widely used in search engines like Google
- Provides accurate ranking of documents

Example: If a document and a query share many important terms (with high weights), the angle between their vectors will be small, and cosine similarity will be high. Hence, that document will rank higher in search results.

The Vector Space Model represents documents and queries as vectors, enabling mathematical comparison. Cosine similarity is then used to measure how close these vectors are, helping in ranking documents based on relevance. This combination forms a fundamental technique in modern Information Retrieval systems.

Ques 4. Compare Boolean retrieval model and vector space retrieval model.

In Information Retrieval (IR), both the Boolean Retrieval Model and the Vector Space Model are used to retrieve documents, but they differ significantly in their approach and effectiveness.

1. <u>Boolean Retrieval Model:</u>

The Boolean Retrieval Model is one of the earliest models in IR. It is based on **Boolean logic** using operators such as AND, OR, and NOT.

- Documents are either **relevant or not relevant** (no ranking)
- Queries are expressed using Boolean operators
- Example: *data AND mining* **Characteristics:**
- Exact match required
- No partial matching
- Results are not ranked
- Simple and easy to implement

2. <u>Vector Space Retrieval Model:</u>

The Vector Space Model represents documents and queries as **vectors in a multidimensional space** and measures similarity between them.

- Uses **weights (like TF-IDF)** for terms
- Supports **partial matching**
- Documents are **ranked based on similarity**

Characteristics:

- Allows degree of relevance
- Uses similarity measures (like cosine similarity)
- Provides ranked results
- More flexible and effective **Comparison Table:**

Feature	**Boolean Retrieval Model**	**Vector Space Model**
Basis	Boolean logic (AND, OR, NOT)	Vector representation
Matching	Exact match	Partial match allowed
Ranking	No ranking	Ranked results
Result Type	Relevant / Not relevant	Degree of relevance

Query Type	Complex Boolean expressions	Simple keyword queries
Flexibility	Less flexible	More flexible
Efficiency	Fast but limited	More accurate but computationally heavier

Key Difference: The Boolean model gives **binary results (yes/no)**, while the Vector Space Model provides **graded relevance (how much relevant)**.

While the Boolean Retrieval Model is simple and useful for precise queries, the Vector Space Model is more powerful and widely used because it ranks documents and handles partial matches effectively. Modern IR systems mainly rely on vectorbased approaches for better performance.

Unit III – Text Categorization and Filtering

Ques 1. Explain the concept of text classification and its applications in spam filtering.

Text classification is the process of automatically assigning predefined categories or labels to text documents based on their content. It is a key technique in Information Retrieval and Natural Language Processing used to organize and manage large amounts of textual data.

In simple terms, it means sorting text into groups such as spam/not spam, positive/negative, news categories, etc.

How Text Classification Works:

1. Data Collection:
   A set of documents is collected (emails, messages, articles, etc.)
2. Preprocessing:
   Text is cleaned by removing stop words, punctuation, and converting to lowercase.

3. Feature Extraction:
   Important words or features are extracted using techniques like TF-IDF.
4. Model Training:
   A classification model (like Naive Bayes, Decision Tree, etc.) is trained using labeled data.
5. Classification:
   The trained model assigns a category to new, unseen text.

Applications of Text Classification:

- Spam detection in emails
- Sentiment analysis (positive/negative reviews)
- News categorization
- Document organization
- Language detection

Spam Filtering using Text Classification:

Spam filtering is one of the most common and important applications of text classification. It is used to automatically classify emails as spam (unwanted) or not spam (important).

Email services like Gmail use advanced text classification techniques to filter spam messages.

How Spam Filtering Works:

1. Training Data:
   The system is trained using a dataset of emails labeled as spam or non-spam.
2. Feature Identification:
   Certain words like “free”, “win”, “offer”, “lottery” are often associated with spam emails.
3. Probability Calculation (e.g., Naive Bayes):

The system calculates the probability that an email is spam based on the words it contains.

4. Classification:
   If the probability exceeds a certain threshold, the email is marked as spam.

Benefits of Spam Filtering:

- Protects users from unwanted emails
- Saves time by reducing clutter
- Improves email security
- Enhances user experience

Text classification is an essential technique for organizing and analyzing textual data. Its application in spam filtering helps in automatically detecting and removing unwanted emails, making communication more efficient and secure.

Ques 2. Describe the Naïve Bayes model used for text classification.

The Naïve Bayes model is a probabilistic classification algorithm based on Bayes' Theorem. It is widely used in text classification tasks such as spam filtering, sentiment analysis, and document categorization.

It is called “naïve” because it assumes that all features (words) in a document are independent of each other, which is a simplifying assumption.

<u>Bayes’ Theorem:</u>

The basic formula used in the Naïve Bayes model is:

$$P(C|X) = \frac{P(X|C)\,P(C)}{P(X)}$$

$P(A)$	0.20
$P(B \mid A)$	0.85
$P(B \mid \neg A)$	0.10

$$P(A \mid B) = \frac{P(B|A)P(A)}{P(B)} \approx 0.68, \;\; P(B) \approx 0.25$$

Where:

- $P(C \mid X)$: Probability of class Cgiven document X
- $P(X \mid C)$: Probability of document Xgiven class C
- $P(C)$: Prior probability of class
- $P(X)$: Probability of document

How Naïve Bayes Works in Text Classification:

1. **<u>Training Phase</u>:**
    - The model is trained on a labelled dataset (e.g., spam and non-spam emails).
    - It calculates the probability of each word occurring in each class.
2. **<u>Feature Independence Assumption:</u>**
    - Words are treated as independent features.
    - Example: The words "free" and "offer" are considered independent even if they often appear together.
3. **<u>Probability Calculation:</u>**
    - For a new document, the probability of each class is calculated using Bayes' theorem.
    - The model computes:

$$P(C \mid X) \propto P(C) \times \prod P(w_i \mid C)$$

4. **<u>Classification:</u>**
    - The document is assigned to the class with the **highest probability**.

Example: If an email contains words like "win", "free", and "offer", the probability of it being spam becomes higher based on previously learned data.

<u>Advantages of Naïve Bayes:</u>

- Simple and easy to implement
- Works well with large text data
- Fast and efficient
- Performs well even with limited training data

Limitations:

- Assumes independence of features (which is not always true) • May be less accurate compared to more complex models

The Naïve Bayes model is a powerful and efficient method for text classification. Despite its simplifying assumptions, it performs remarkably well in real-world applications like spam filtering and is widely used in Information Retrieval systems.

Ques 3. Explain the working principle of k-Nearest Neighbors for document classification.

k-Nearest Neighbors (k-NN) is a **supervised learning algorithm** used for classification and prediction. In document classification, it classifies a new document based on the categories of its **nearest (most similar) documents** in the training dataset.

It is called a **lazy learning algorithm** because it does not build a model during training; instead, it stores all the training data and performs computation at the time of classification.

Working Principle of k-NN:

1. **Training Phase:**
    - All documents in the dataset are stored along with their class labels (e.g., spam or non-spam). ○ No explicit model is created.
2. **Representation of Documents:**

- Each document is converted into a **vector form** using techniques like TF-IDF.

3. **<u>Distance/Similarity Calculation:</u>**
   - When a new document is given, its similarity (or distance) is calculated with all training documents.
   - Common measures include:
     - Cosine similarity
     - Euclidean distance
4. **<u>Finding k Nearest Neighbors:</u>**
   - The system selects the top **k documents** that are most similar to the new document.
5. **<u>Majority Voting:</u>**
   - The class of the new document is determined by the **majority class** among the k nearest neighbors.
6. **<u>Final Classification:</u>**
   - The document is assigned to the category that appears most frequently among its neighbors.

Example:

If $k = 3$and among the 3 nearest documents:

- 2 belong to "Sports"
- 1 belongs to "Politics"

Then the new document will be classified as **"Sports"**.

<u>Advantages of k-NN:</u>

- Simple and easy to understand
- No training phase required
- Works well with multi-class problems

Limitations:

- Computationally expensive (needs to compare with all documents)
- Sensitive to the choice of k
- Performance depends on distance/similarity measure

The k-Nearest Neighbors algorithm classifies documents based on similarity to existing labeled documents. By using distance measures and majority voting, it effectively assigns categories, making it a useful technique in text classification tasks.

Ques 4. Compare Support Vector Machine classifiers with centroid-based classifiers in text categorization.

In text categorization, different machine learning techniques are used to classify documents into categories. Two important approaches are **Support Vector Machine (SVM) classifiers** and **centroid-based classifiers**. They differ in their working principles, complexity, and performance.

1. Support Vector Machine (SVM) Classifier:

A Support Vector Machine is a supervised learning algorithm that finds an **optimal hyperplane** to separate data into different classes.

Key Features:

- Maximizes the margin between classes
- Works well in **high-dimensional spaces** (like text data)
- Uses support vectors (critical data points near boundary)
- Can handle linear and non-linear classification (using kernels)

Working Idea:
It tries to draw a boundary that best separates documents of different categories with maximum distance from nearest points.

2. Centroid-Based Classifier:

A centroid-based classifier represents each class by a **centroid (average vector)** of all documents belonging to that class.

Key Features:

- Computes the average vector for each category
- Classifies a new document based on **similarity to centroids**
- Simple and fast
- Often uses cosine similarity

Working Idea:
A document is assigned to the class whose centroid is closest (most similar) to it.

Comparison Table:

Feature	SVM Classifier	Centroid-Based Classifier
Approach	Boundary-based (hyperplane)	Similarity-based (centroid)
Complexity	High	Low
Training Time	Longer	Faster
Accuracy	High	Moderate
Handling High Dimensions	Very effective	Effective but simpler
Model Type	Discriminative	Prototype-based
Flexibility	Can handle comp	Limited flexibility

Key Differences:

- SVM focuses on **finding a decision boundary**, while centroid-based methods focus on **finding the nearest class center**
- SVM generally provides **higher accuracy**, especially for complex datasets
- Centroid-based classifiers are **simpler and faster**, making them suitable for large datasets with limited resources

Both SVM and centroid-based classifiers are useful in text categorization. SVM is preferred when high accuracy is required, while centroid-based classifiers are useful when simplicity and speed are important. The choice depends on the application and computational resources.